

Implementation Plan - Backend Validation Integration & Documentation Update

Align the frontend components, form validations, error handling, and documentation with the backend validation changes committed in `3881edb` (Server Validation).

User Review Required

> [!IMPORTANT]

> - **Gender Case Sensitivity**: Backend `@Pattern(regex = "Male|Female|Other")` is case-sensitive. The frontend `Profile.jsx` select element used `MALE`, `FEMALE`, `OTHER`, which will be updated to `Male`, `Female`, `Other`.

> - **Missing Email Field**: `CreateEmployee.jsx` was missing the `email` field required by backend's `EmployeeRequest` (`@NotBlank`, `@Email`). This field will be added to the UI.

> - **Account Opening Page**: `Accounts.jsx` was empty and missing from `App.jsx` routes despite being linked from `CustomerDashboard.jsx`. We will implement this with `initialDeposit` (Min 1000) validation.

Proposed Changes

Frontend Error Utility & API Service

[NEW] [errorHandler.js](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/Utils/errorHandler.js)

- Utility function to extract field-level or text validation messages from Spring Boot backend error responses.

[NEW] [accountService.js](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/api/accountService.js)

- Axios API helper for Account Service running on `http://localhost:8084` (`/accounts/open`, `/accounts/me`).

Frontend Components & Pages

[MODIFY] [CreateEmployee.jsx](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/pages/admin/CreateEmployee.jsx)

- Add `email` state and input field to match `EmployeeRequest` DTO.

- Enforce validation rules: `username` (4-20 chars), `password` (min 6 chars), `email` (valid email format).

- Render field errors and backend validation response messages.

[MODIFY] [RegisterCustomer.jsx](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/pages/employee/RegisterCustomer.jsx)

- Enforce validation rules: `username` (4-20 chars), `password` (min 6 chars).
- Render field errors and backend validation error feedback.

[MODIFY] [Profile.jsx](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/pages/customer/Profile.jsx)

- Change dropdown values from `MALE`/`FEMALE`/`OTHER` to `Male`/`Female`/`Other`.
- Add client-side validation:
 - `firstName` & `lastName`: 2-30 chars, letters only (`^[A-Za-z ]+\$`).
 - `email`: Valid email format.
 - `mobile`: 10 digits starting with 6-9 (`^[6-9]\d{9}\$`).
 - `dateOfBirth`: Required, past date, age must be ≥ 18 years.
 - `gender`: Required (`Male`, `Female`, `Other`).
- Render inline field error messages.

[MODIFY] [Kyc.jsx](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/pages/customer/Kyc.jsx)

- Add client-side validation:
 - `aadhaarNumber`: Exactly 12 digits (`^\d{12}\$`).
 - `panNumber`: 10-char uppercase PAN format (`^[A-Z]{5}[0-9]{4}[A-Z]\$`).
 - `address`: 10-250 characters.
- Auto-uppercase PAN input.
- Render inline field error messages.

[MODIFY] [Accounts.jsx](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/pages/customer/Accounts.jsx)

- Build Open Account component with `initialDeposit` validation (min 1000, required).
- Show list of user accounts returned from `GET /accounts/me`.

[MODIFY]

[App.jsx](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/src/App.jsx)

- Register route `/customer/accounts` linking to `Accounts.jsx`.

Documentation

[NEW] [FRONTEND_API_DOCUMENTATION.md](file:///d:/CDAC/Project/Banking-Project/FRONTEND_API_DOCUMENTATION.md)

- Complete, updated API documentation covering all four backend services (Auth Service, Customer Profile Service, KYC Service, Account Service).
- Detailed tables specifying Endpoint, Method, Description, Allowed Roles, Request Body structure, and Validation Rules.

[MODIFY] [README.md](file:///d:/CDAC/Project/Banking-Project/ProjectFrontend/bank-frontend/README.md)

- Update frontend README with backend service ports, validation constraints, and page instructions.

Verification Plan

Automated Tests / Lint

- Check JavaScript files for syntax errors using oxlint / build test:

```
```powershell
cd d:\CDAC\Project\Banking-Project\ProjectFrontend\bank-frontend
npm run build
```
```

Manual Verification

- Test validation rules on `CreateEmployee`, `RegisterCustomer`, `Profile`, `Kyc`, and `Accounts` forms.
- Verify error messages display when submitting invalid data (e.g., age < 18, bad PAN format, initial deposit < 1000).
- Confirm documentation `.md` renders cleanly with complete validation specifications.